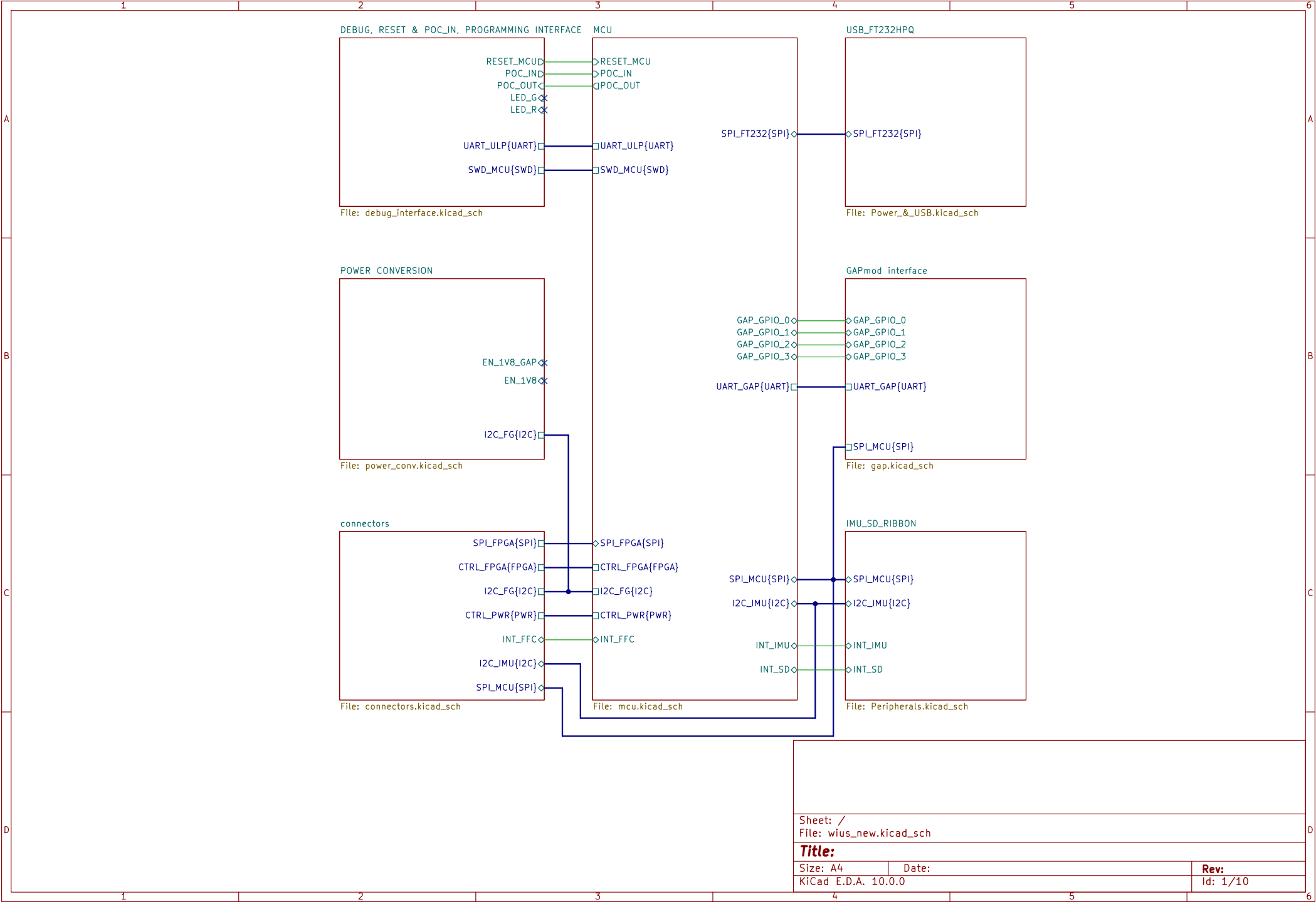
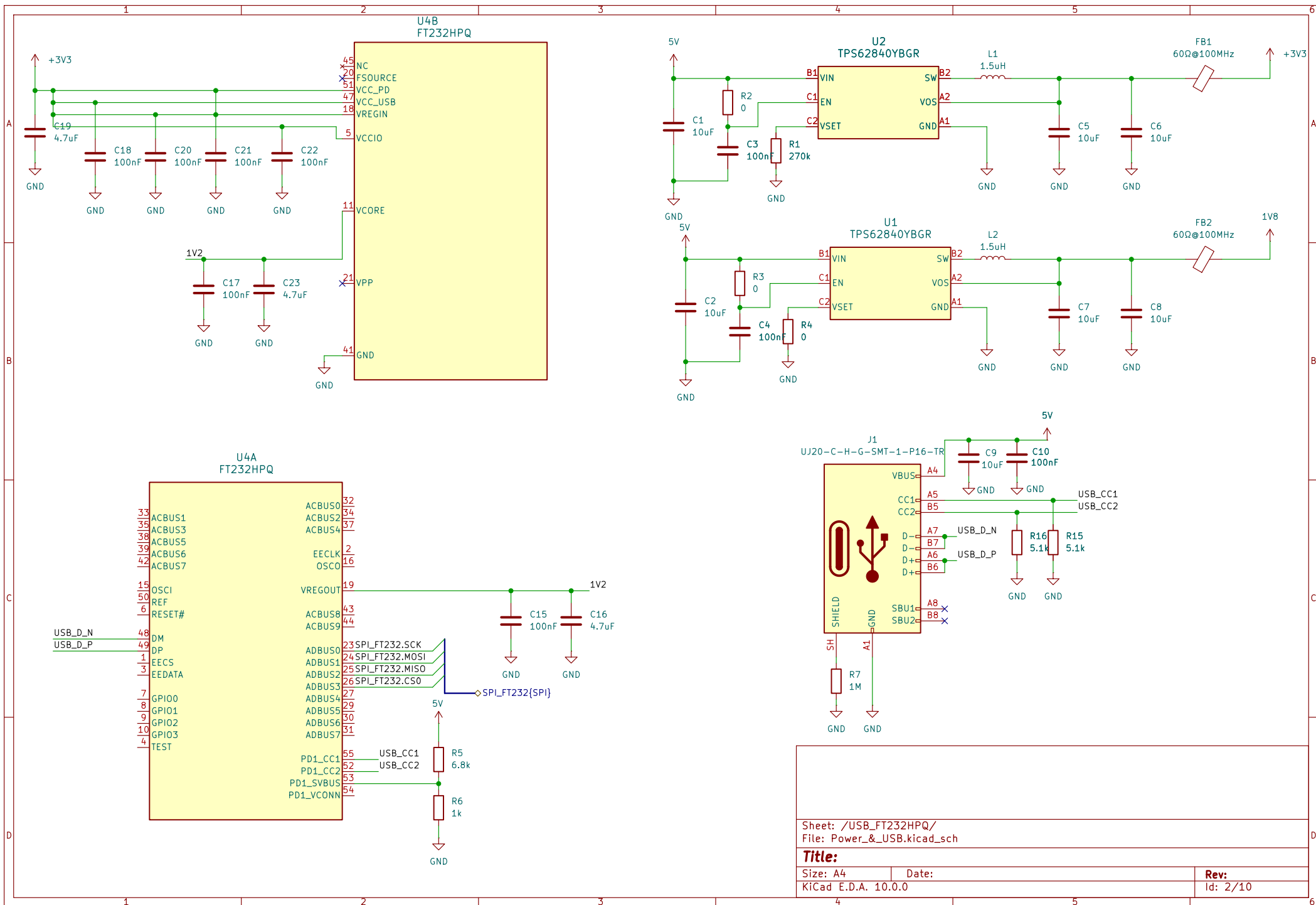






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File: Power_&_USB.kicad_sch

Title:

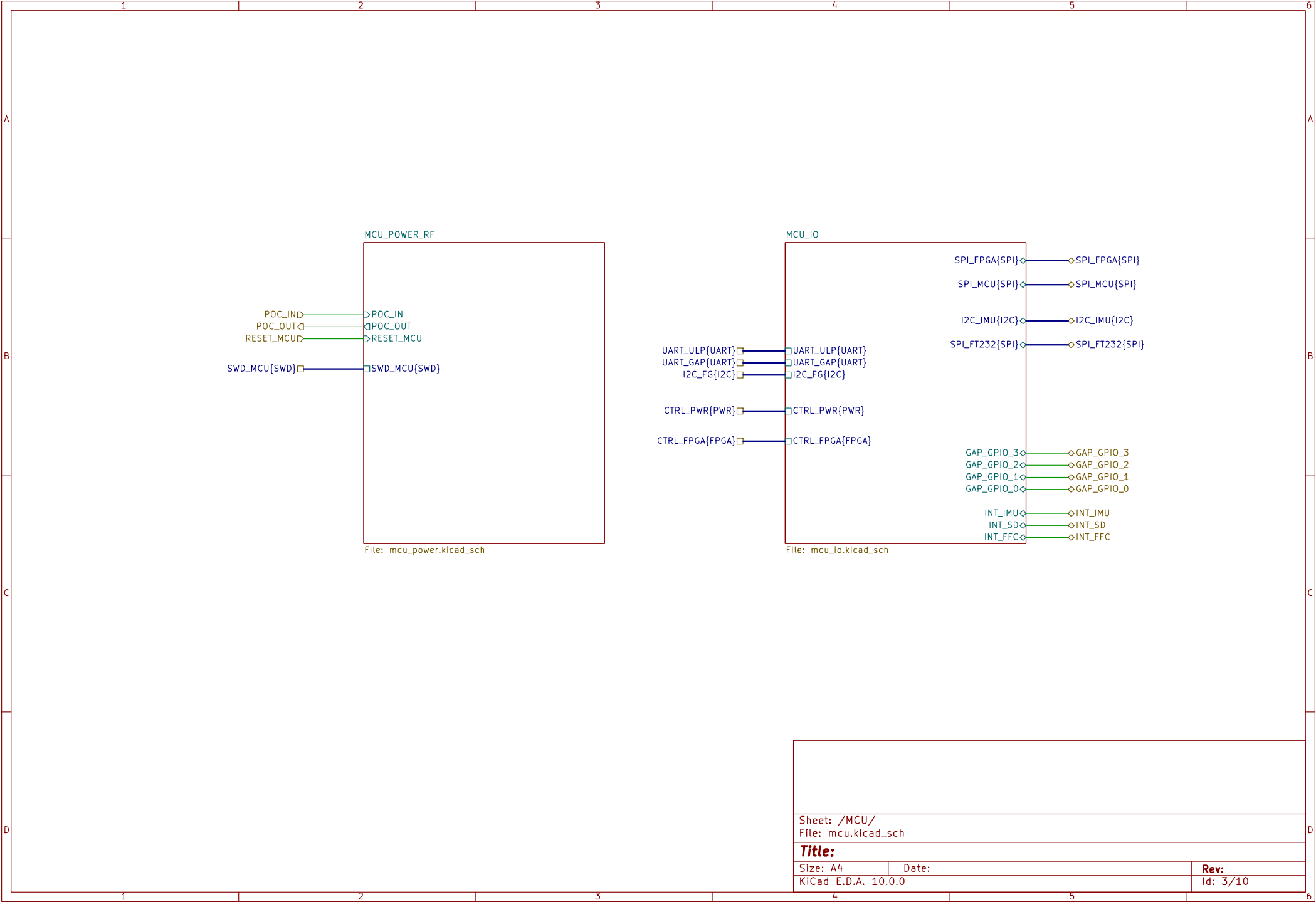
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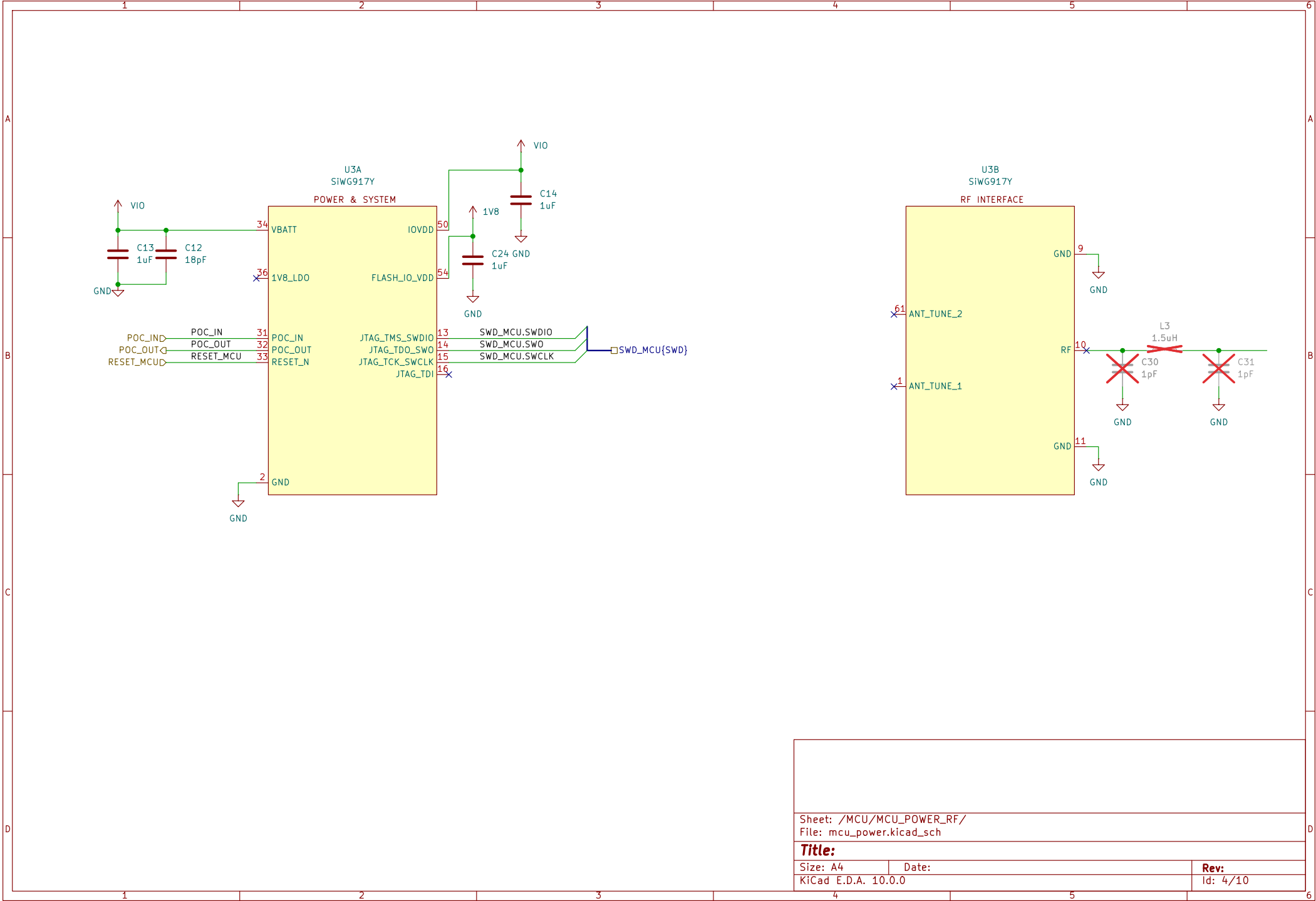
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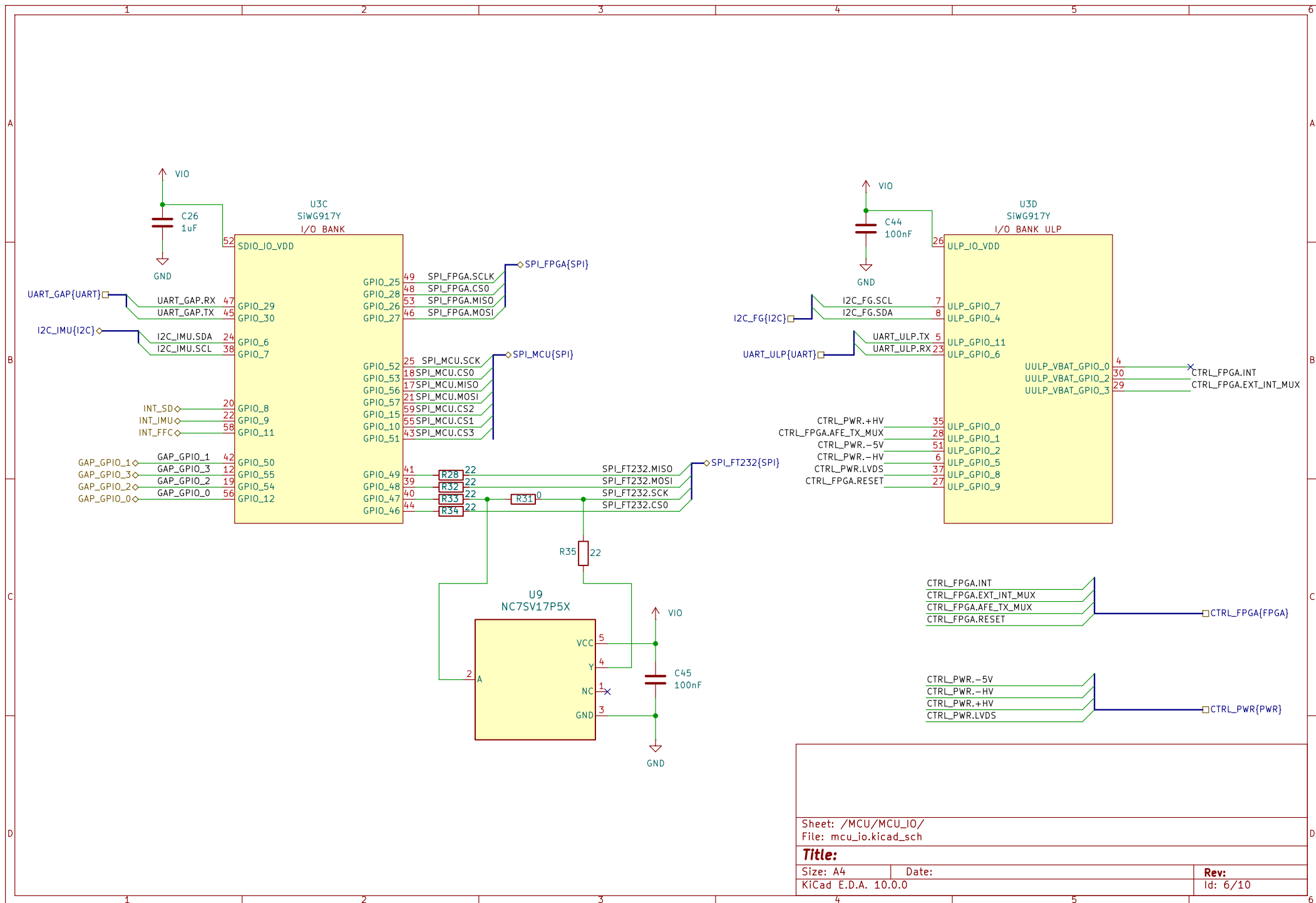
KiCad E.D.A. 10.0.0

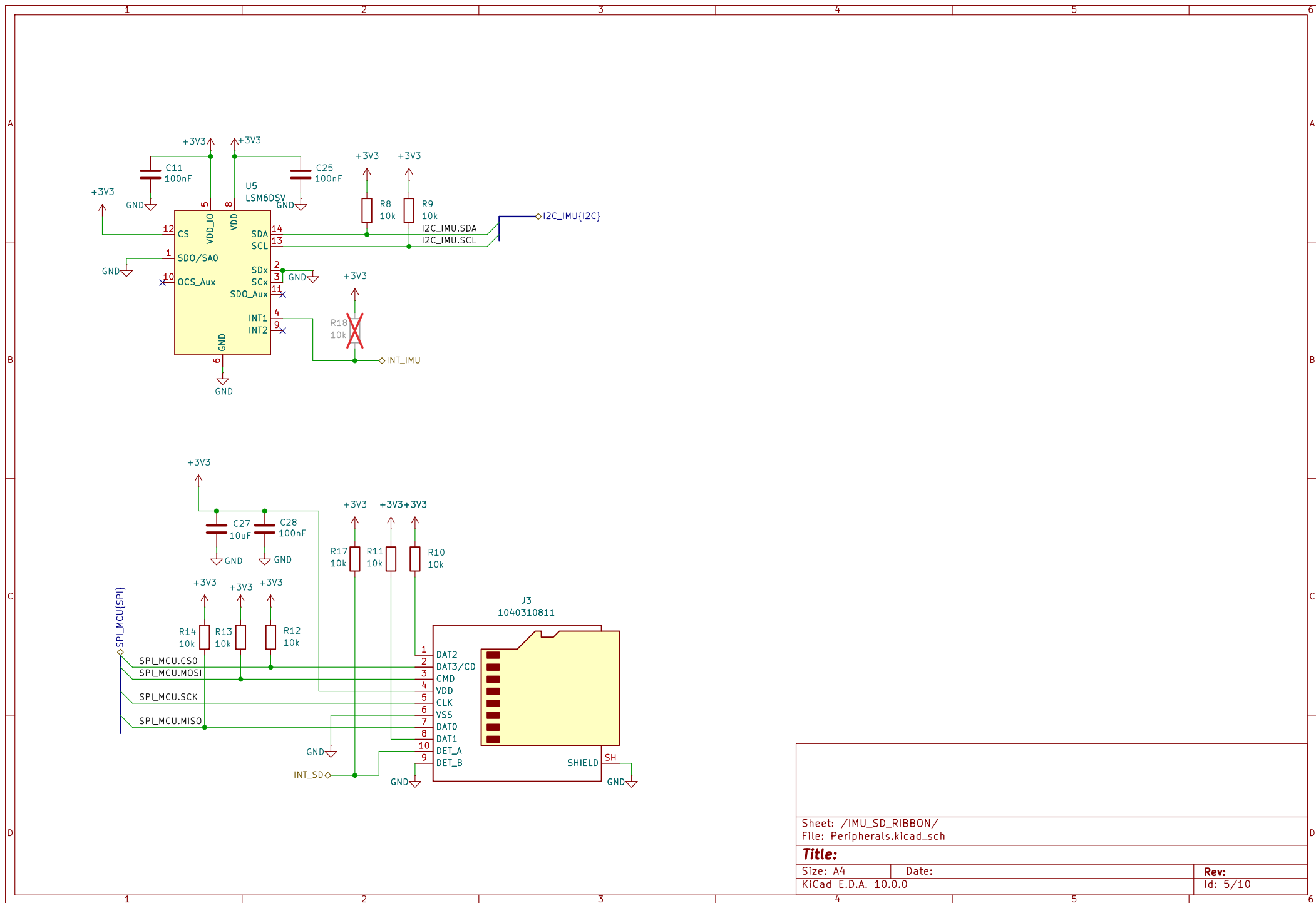
Rev:

Id: 2/10









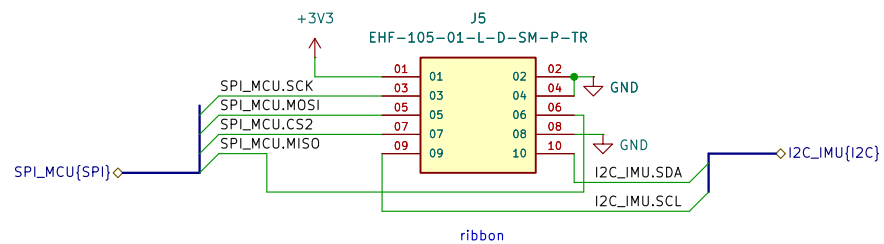
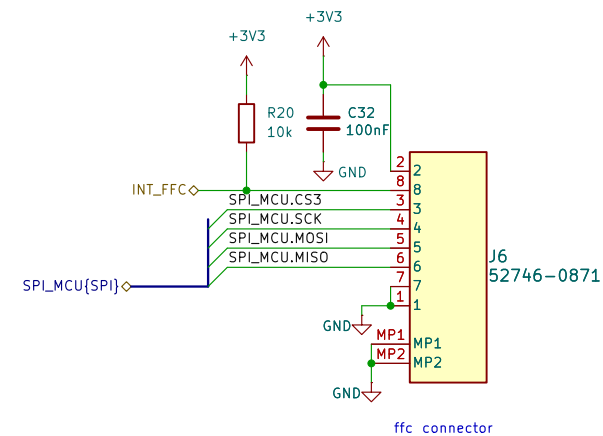
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Size: A4
KiCad E.D.A. 10.0.0

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Id: 5/10



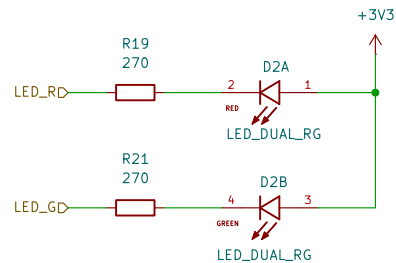
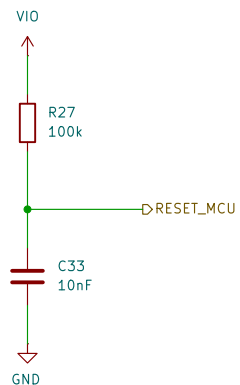
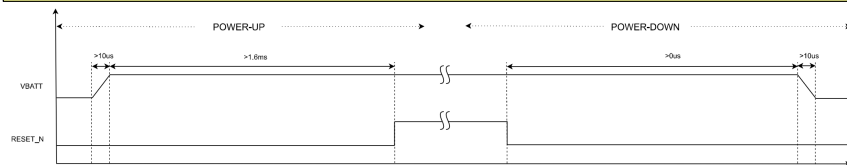
Rev:
Id: 7/10

The IC generates a POC (Power On Control) signal – POC_OUT that is distributed to all I/O cells to prevent the I/O cells from powering up in an undesired configuration and is also used inside the IC to safe state the IC till a valid supply is available for proper operation. This power management is functional in both power up and power down sequences.

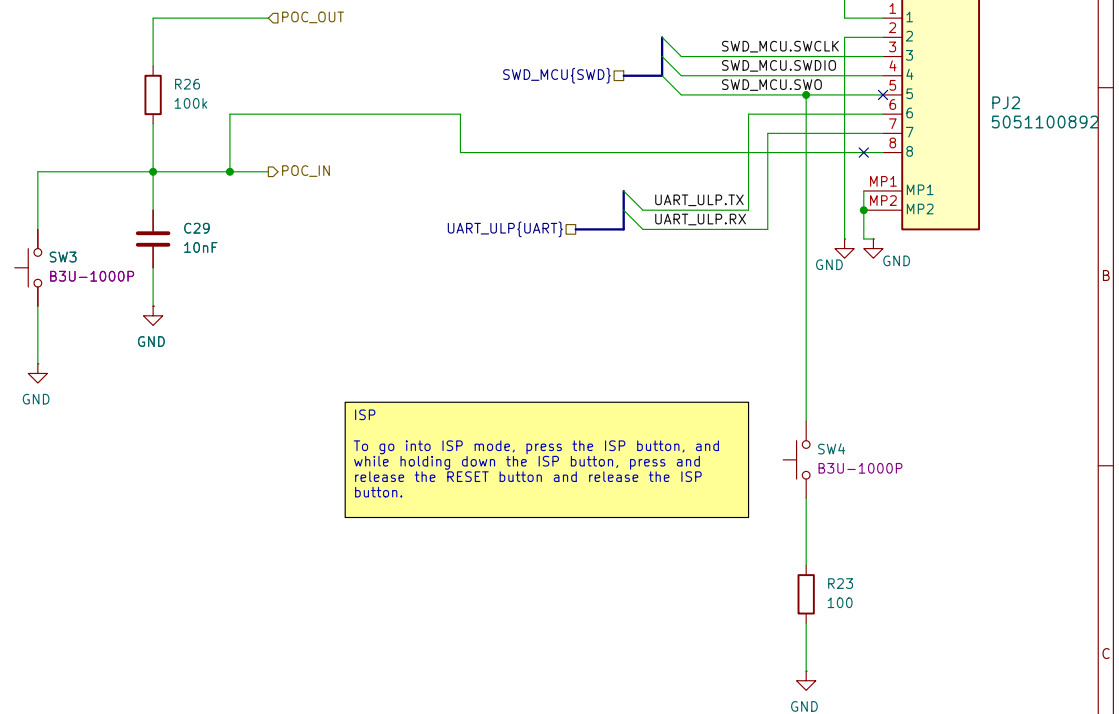
During power up, until the UULP_VBATT_1 and UULP_VBATT_2 (VBATT supply) reach 1.6V, the POC_OUT signal stays low. Once the VBATT supply exceeds 1.6V, the POC_OUT becomes high and normal operation of the IC starts.

Once the POC_OUT becomes high, it stays high. But if VBATT becomes lower than the Blackout threshold voltage, POC_OUT becomes low.

The following figure illustrates the power up sequence when POC_OUT is connected to POC_IN. As shown in the figure below, the RESET_N is high at least 1.6ms after VBATT supply is stable. The RESET_N signal can be controlled via options like an R/C circuit or another MCU's GPIO.



Vdd = 3.3V
If = 5mA for both
Vf,red = 1.84V
Vf,green = 1.96V
--> R,red = 2920hm
R,green = 2680hm



Sheet: /DEBUG, RESET & POC_IN, PROGRAMMING INTERFACE/
File: debug_interface.kicad_sch

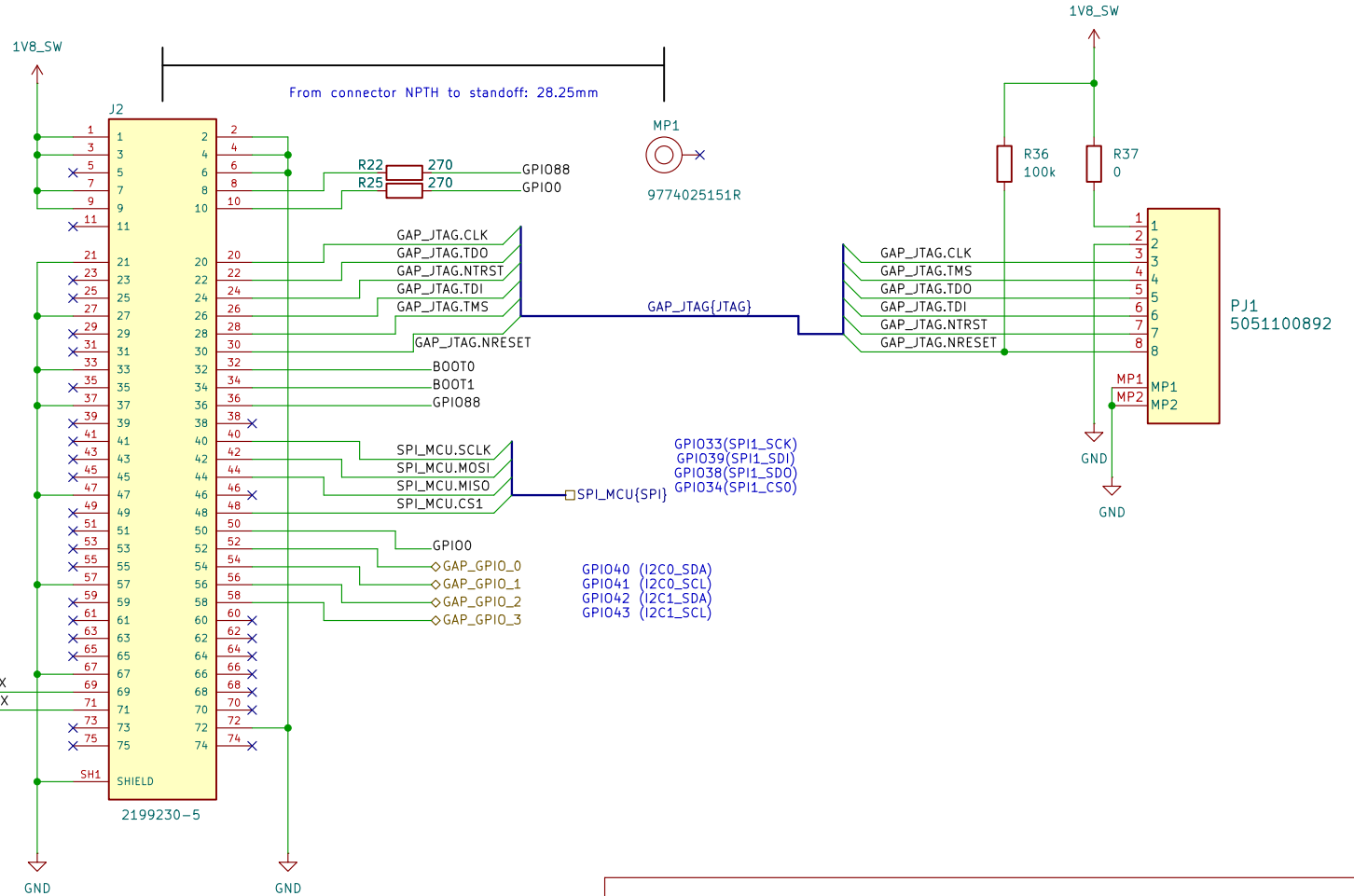
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Rev:
Id: 8/10

BOOT0/BOOT1: Boot source

0/0: as per eFuse BOOTMODE
0/1: JTAG
1/0: Hyperflash
1/1: eMRAM (default)



Sheet: /GAPmod interface/
File: gap.kicad_sch

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Rev:
Id: 9/10

